



Mood Light

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A diffused ambient light built from repurposed materials, controlled directly from any modern web browser using the Web Bluetooth API.

(A photo of the mood light in action, glowing, and maybe a shot of the 3D model in Onshape would be great here.)

The Story

This project was born from a happy accident. I had some leftover WS2811 LED strips from my thesis project and a few plexiglass samples that I had originally ordered for the NetScore project—I thought they'd be clear display protectors, but they turned out to be translucent. This "mistake," however, turned out to be the perfect material for a light diffuser.

I designed a simple enclosure in Onshape, consisting of a 3D-printed frame that wraps around the plexiglass panel, with the LED strip running along the inner perimeter. The result is a soft, almost uniform glow.

The light is powered by an ESP32-C3, and the LEDs are controlled wirelessly via a web app using the Web Bluetooth API, allowing you to connect and change colors on the fly from a browser.

Key Features

- **Soft, Diffused Lighting:** Uses a WS2811 LED strip paired with a translucent plexiglass panel for a smooth, ambient glow.
- **Custom Enclosure:** The frame was parametrically designed in Onshape.
- **Browser-Based Control:** Wirelessly change colors and patterns using the Web Bluetooth API—no native app required.
- **Repurposed Components:** Built entirely from leftover materials from previous projects.

Tech Stack

- **Hardware:** ESP32-C3, WS2811 individually addressable LED strip.
- **Control Protocol:** Web Bluetooth API.
- **3D Design:** Onshape.
- **Circuit Board Technique:** Perfboard for now.

Challenges & Lessons Learned

The primary lesson from this project was the value of creative repurposing and "happy accidents." The translucent plexiglass, initially a purchasing mistake for another project, became the key component that defined the aesthetic and also the driving force in this project's creation.

It's a good reminder that sometimes the best ideas come from adapting to unexpected materials and finding potential in leftovers.

Keywords

3D Printing, Web Bluetooth API, Hardware, Mobile, Lighting, LEDs, Perfboard, Onshape, ESP32, Bluetooth, 12V, 5V